Classical Baseline Report

Detection, identification and characterization of V/UHF signals with classical / statistical methods

EEE493 Senior Design · Bilkent University · Industry partner: TÜBİTAK BİLGEM İLTAREN · Prepared by Cem Palabıyık · 15 October 2026

Summary

- Every classical method we know of for the three pipeline stages was implemented and tested on the same data with the same scoring: **14 detectors**, **8 classifiers plus the likelihood-ratio bound**, **carrier, bandwidth, SNR and symbol-rate estimators**, plus waveform-specific and auxiliary methods (matched filter, OFDM cyclic-prefix detector, MDL/AIC, CUSUM, wavelet edges, Hough transform). Each was checked against theory or an independent re-implementation.
- **Detection:** an energy detector with a gain(t) × shape(f) noise-floor model is best (box F1 67 % clean, 53 % with receiver impairments). Low SNR (0–5 dB) is the weak spot: at most 40–71 % of signals found.
- **Identification:** on our 7 target classes, feature-based classifiers reach ~100 % above 10 dB and 85 % at 2 dB; on all 24 RadioML classes no classical method reaches 85 % (best 63 %). A practical likelihood classifier (all parameters estimated blindly) beats the features when the symbol rate is right (RadioML 85 % at 2–4 dB vs 6 dB), at ~800× the cost.
- **Characterization:** carrier to 0.4 % of the bandwidth, SNR to 0.45 dB median (clean); symbol rate within 2 % for 79 % of frames.
- **All results are on synthetic data** (RadioML 2018.01A and TorchSig scenes). No real SDR recordings yet.

1. Data and evaluation rules

Stage	Data	What is scored
Detection (WP4)	600 TorchSig 2.2 wideband scenes (WidebandSig53-style): 300 with level-2 impairments, 300 clean. 262,144 samples at 10 MS/s (512 × 512 spectrogram), 1–6 emitters per scene from 69 signal types, SNR −5…30 dB.	Box precision / recall / F1 (IoU ≥ 0.5, Hungarian matching), emitter Pd, measured false-alarm rate, recall vs SNR
Identification (WP5)	RadioML 2018.01A: 7-class subset (AM-DSB-WC, FM, BPSK, QPSK, 8PSK, 16QAM, GMSK) and all 24 classes, −20…30 dB. Symbol-level synthetic frames for the likelihood bound.	Accuracy overall, at SNR ≥ 10 dB, and per SNR
Characterization (WP6)	Same TorchSig scenes (truth boxes give carrier, bandwidth, SNR). Synthetic RRC-shaped PSK/QAM and GMSK for symbol rate (no symbol-rate labels in TorchSig).	Median and 90th-percentile error; success within 2 % for symbol rate

- Tuning only on scenes 0–99 or the training split; **all reported numbers are on held-out scenes 100–299 / the test split**.
- Every detector's threshold is calibrated on pure white noise, so a target false-alarm rate of 10^{-3} means the same for all of them.
- RadioML labels follow the class order in the RadioML 2018 paper. The *classes.txt* shipped with the dataset is wrong for the HDF5 file (verified on all 24 classes).

2. Detection (WP4): 14 detectors

Each detector produces a test-statistic map on the 512 × 512 spectrogram grid. The same segmentation (job 4.3: morphological closing/opening, connected components, minimum 120 cells) turns every mask into boxes, and the same scoring compares them. Numbers at target Pfa 10^{-3}, scenes 100–299.

Impaired scenes (TorchSig level 2)

Detector	Box F1	Precision	Recall	Recall 0–5 dB	Recall 10–15 dB	ms / scene
Energy (excision noise floor)	53.0	48.9	57.8	40	79	318
Energy, multitaper	42.7	34.1	56.9	38	79	406
CFAR, ordered statistic	39.8	36.7	43.4	33	71	209
FCME	35.6	36.9	34.5	16	51	92
CFAR, greatest-of	34.5	33.9	35.2	24	60	20
LAD (double threshold)	30.5	20.7	57.5	38	78	366
CFAR, 2-D	28.8	22.2	41.0	27	73	14
CFAR, cell-averaging	28.6	21.6	42.3	31	69	20
CFAR, smallest-of	15.6	9.9	36.7	26	51	20
Anderson–Darling GoF *	13.7	11.7	16.6	8	23	28
Cyclostationary *	12.3	9.2	18.4	2	21	106
PSD occupancy *	9.9	6.1	26.0	17	34	98
Spectral kurtosis	5.8	6.2	5.5	2	9	18
Eigenvalue (MME) *	4.7	3.4	7.4	4	10	298

Clean scenes

Detector	Box F1	Precision	Recall	Recall 0–5 dB	Recall 10–15 dB
Energy (excision noise floor)	67.4	72.7	62.8	52	74
LAD (double threshold)	66.6	68.1	65.2	57	79
FCME	65.1	71.8	59.5	40	70
CFAR, smallest-of	64.1	60.3	68.5	71	77
Energy, multitaper	60.7	58.3	63.3	58	77
CFAR, cell-averaging	52.1	52.5	51.7	59	72
CFAR, ordered statistic	48.8	47.8	49.8	61	60
CFAR, 2-D	45.6	43.6	47.7	49	70
CFAR, greatest-of	41.2	41.9	40.5	51	47
Eigenvalue (MME) *	37.5	35.4	39.9	26	50
Anderson–Darling GoF *	36.2	31.2	43.0	30	51
PSD occupancy *	23.8	27.1	21.3	16	22
Cyclostationary *	19.6	19.0	20.2	3	18
Spectral kurtosis	6.3	7.3	5.5	3	2

* Decides per block (16 frames × 32 bins) or per whole frequency column, so its boxes are coarse by construction; compare these on the operating curves (Figure 2) rather than on box F1.

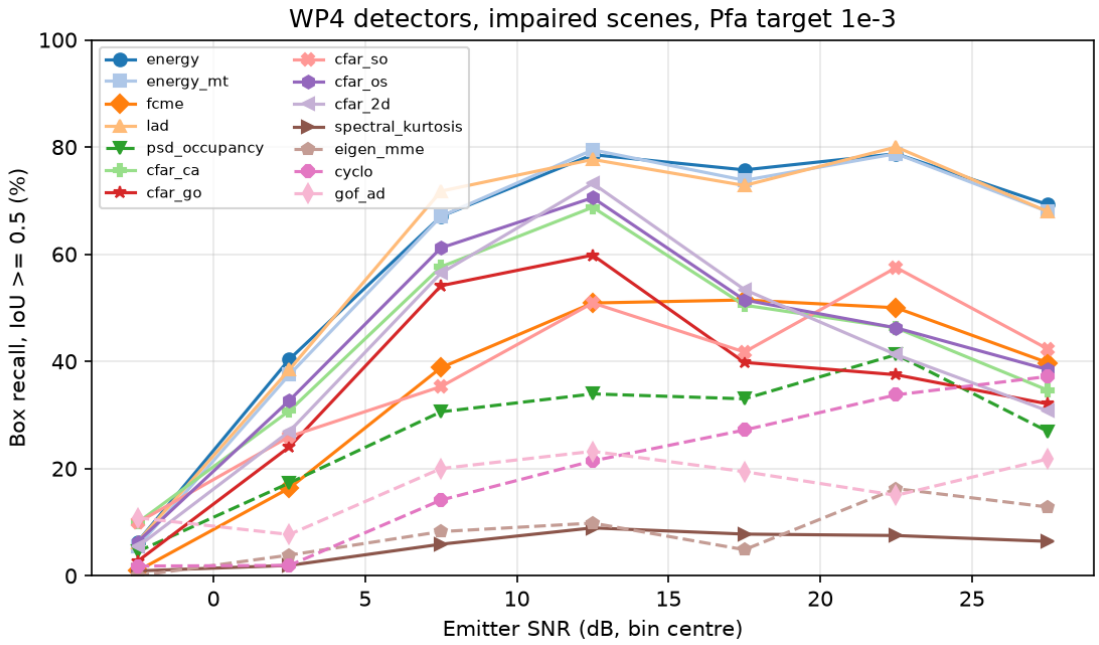


Figure 1. Box recall vs emitter SNR on impaired scenes, all 14 detectors (target Pfa 10^{-3}). Dashed = coarse/block detectors.

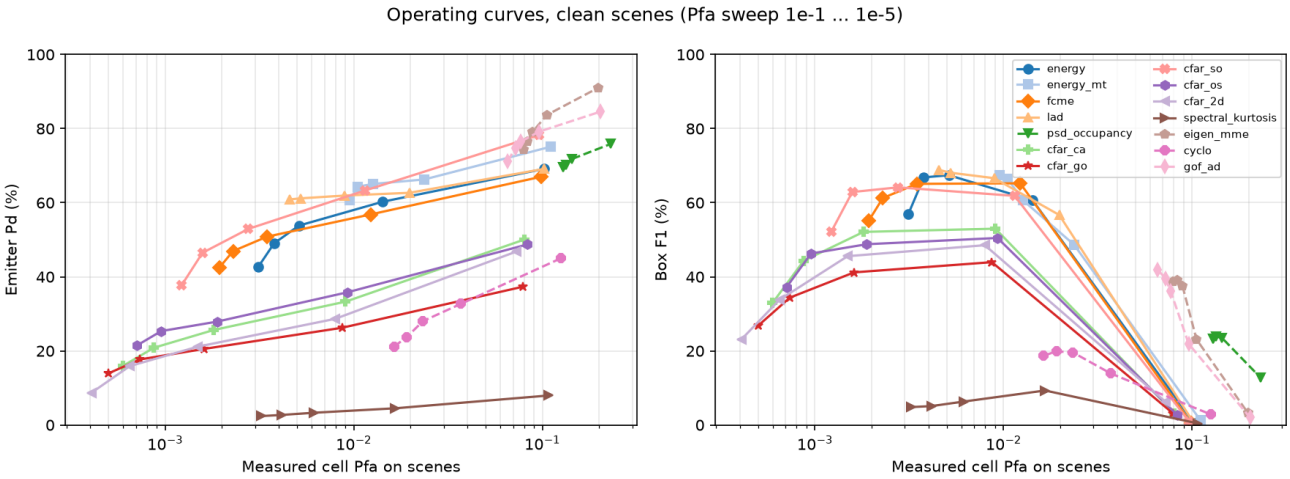


Figure 2. Operating curves on clean scenes: emitter Pd and box F1 against the false-alarm rate actually measured on scenes, over a target sweep 10^{-1} … 10^{-5}. This is the fair comparison across detector types.

Findings

- **The noise-floor model matters more than the test.** One global noise level gave a measured false-alarm rate of 0.28 instead of 10^{-3} on impaired scenes (curved receiver passband, gain steps, AGC). Modelling noise as gain(t) × shape(f) and excluding signal blobs from the estimate fixed it.
- **Impairments cost about 15 F1 points** (energy 67 → 53). On pure noise every detector meets its target exactly; the extra detections on scenes are mostly real but unlabelled energy (IQ-imbalance images, intermodulation, DC spike) and gain steps that change the noise shape mid-scene.
- **1-D CFAR masks wide signals.** With guard 4 / train 16 it found 5 % of emitters, because emitters are 5–64 bins wide and the training cells fall inside the signal. Guard 16 / train 32 (tuned on dev scenes) gives 25–57 % emitter Pd. Recall also falls above 15 dB, where strong signal skirts leak into the training cells. Smallest-of CFAR is the best CFAR on clean data, ordered-statistic on impaired data.
- Eigenvalue and goodness-of-fit tests assume white noise; the receiver passband makes the noise correlated, so on impaired data they fire almost everywhere. Spectral kurtosis only sees non-Gaussian signals and TorchSig's dense signals look Gaussian. Cyclostationary features miss constant-envelope signals without x^2 lines.
- **Low SNR is where AI detection (job 4.4) has to win:** at 0–5 dB the best classical detectors find 40–71 % of signals.

3. Identification (WP5)

All classifiers use the same 19 standard statistical features (higher-order cumulants, power-law spectral lines, amplitude and instantaneous-frequency statistics, spectrum shape) and the same stratified 75/25 split. Feature extraction costs 0.5–0.6 ms per frame for all of them.

Feature classifiers on RadioML 2018.01A

Model	7 cls: all SNRs	7 cls: SNR ≥ 10	85 % at	24 cls: all	24 cls: SNR ≥ 10	Predict ms/frame
Gradient-boosted trees	71.4	99.95	2 dB	39.2	62.8	0.04
Random forest	71.3	99.97	2 dB	39.9	63.1	0.04
SVM (RBF)	71.3	99.96	2 dB	37.6	60.0	2.7
k-NN (k = 15)	70.4	99.9	2 dB	35.4	56.0	0.13
LDA	68.6	99.5	4 dB	28.7	46.8	0.0001
Decision tree, depth 6 (≈ Azzouz–Nandi)	63.8	99.4	6 dB	24.2	41.1	0.0004
Nearest centroid	48.3	70.4	not held	16.0	23.4	0.0002
Cumulant theory, training-free (Swami–Sadler) †	40.9	54.2	never	11.4	14.0	0.0002

Accuracy in %. 7 classes = AM-DSB-WC, FM, BPSK, QPSK, 8PSK, 16QAM, GMSK. No model reaches 85 % at any SNR on 24 classes. † Scored only on the digital classes it can describe (4 of 7, 12 of 24).

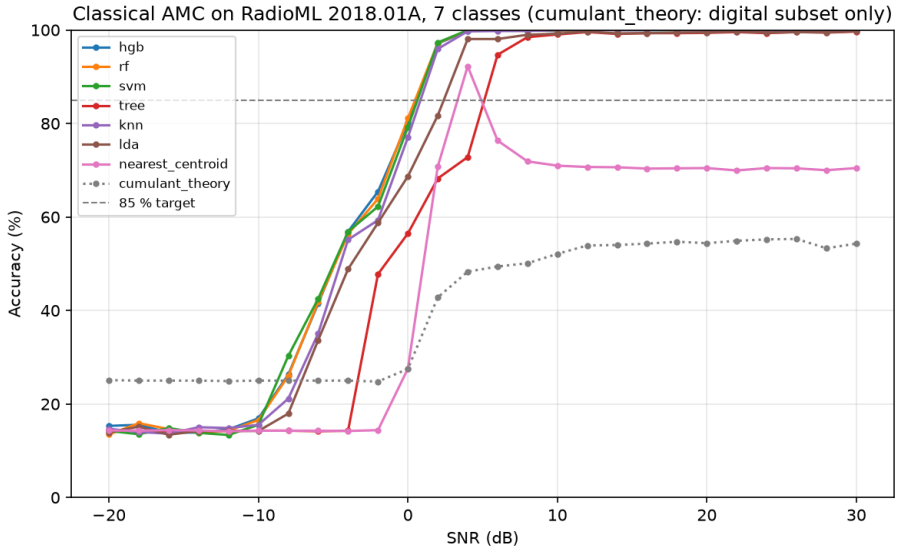


Figure 3. Accuracy vs SNR on the 7-class RadioML subset, all classifiers.

Likelihood-ratio classifier: the classical upper bound

The likelihood classifier is optimal when the receiver knows symbol timing, SNR and the candidate constellations, with only the carrier phase unknown. RadioML does not provide those, so it is tested on symbol-level synthetic frames (BPSK, QPSK, 8PSK, 16QAM, 64QAM; 128 symbols, AWGN, random phase), with the feature classifier trained and tested on the same kind of frames.

Method	All SNRs (−10…20 dB)	SNR ≥ 10 dB	85 % reached at	ms / frame
ALRT (phase-averaged likelihood)	70.2	99.2	8 dB	15
HLRT (phase-maximised likelihood)	69.1	99.1	8 dB	15
19 features + boosted trees	61.1	89.2	14 dB	0.05
Cumulant theory, training-free	45.8	81.1	18 dB	0.05

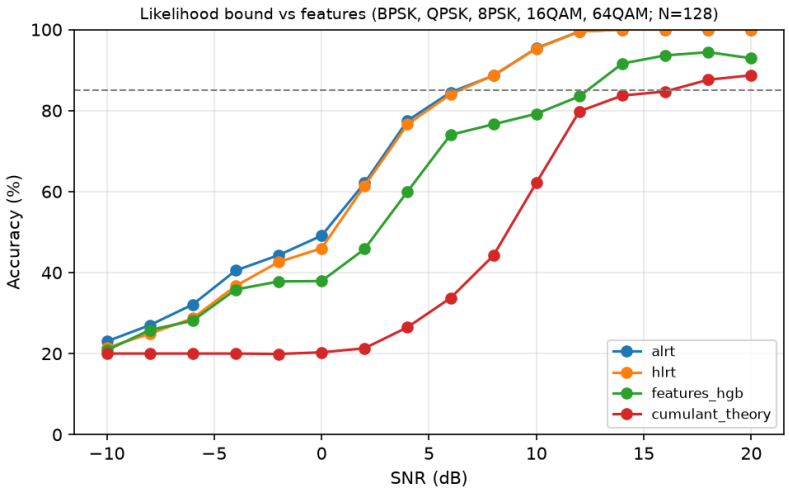


Figure 4. Likelihood-ratio bound vs feature-based classification on identical frames.

Practical likelihood classifier: no oracle knowledge

Every nuisance parameter is estimated per frame: symbol rate (smoothed instantaneous-frequency estimator), resampling to 8 samples/symbol, RRC matched filter (roll-off 0.35 assumed), timing, carrier offset (4th/8th-power line); phase averaged (ALRT) or maximised (HLRT); noise level maximised over a grid. Same test frames for all methods; the feature classifier is trained on the other half. Classes BPSK, QPSK, 8PSK, 16QAM, 64QAM, −10…20 dB. Unit check: 48/50 correct on clean 25 dB frames.

Source	Method	All SNRs	SNR ≥ 10	85 % at	ms / frame
RadioML 2018.01A (4,800 test frames)	19 features + boosted trees	66.2	87.1	6 dB	0.05
	HLRT, symbol rate known (8 sps)	70.6	93.4	2 dB	42
	ALRT, symbol rate known (8 sps)	70.4	93.3	4 dB	42
	ALRT, fully blind	59.0	80.4	never	42
TorchSig modulators (3,200 test frames)	19 features + boosted trees	62.1	85.4	16 dB	0.05
	ALRT, symbol rate known	76.3	95.6	4 dB	38
	ALRT, fully blind	69.1	91.3	6 dB	38
	HLRT, fully blind	69.0	90.9	6 dB	38

With a correct symbol rate the likelihood classifier beats the features on both sources. Fully blind it still wins on TorchSig (by 10 dB) but loses on RadioML, where the symbol-rate estimator mis-reads QAM frames (11–12 instead of 8 samples/symbol). It costs ~800× more time per frame.

Findings

- On 7 well-separated classes the classifier barely matters (boosted trees ≈ random forest ≈ SVM ≈ k-NN): **the features are the limit**. Boosted trees give the best accuracy-to-speed trade-off.
- **On all 24 classes the best is ~63 % at high SNR.** Failures: 8/16/32PSK (identical cumulants), high-order QAM/APSK, AM-SSB with vs without carrier, OOK vs BPSK. The 85 % target is reachable classically only on a reduced, well-separated class set.
- **Feature classifiers are about 6 dB below the likelihood bound** (85 % at 14 dB vs 8 dB): roughly the room an AI model can claim without channel knowledge.
- Theoretical cumulant values do not survive pulse shaping and noise (BPSK |C40| measures 1.45 instead of 2), so training-free classification is weak.

4. Characterization (WP6)

Carrier, bandwidth and SNR are estimated inside each truth box (oracle detection), from a Welch PSD of the box's time span with the noise level taken from both sides of the signal. Errors are median / 90th percentile on scenes 100–299. The best estimator per quantity was chosen on dev scenes.

Quantity	Best, clean	Best, impaired	Other estimators (median, clean / impaired)
Carrier (error ÷ true bandwidth)	Power centroid: 0.004 / 0.088	−10 dB-edge midpoint: 0.015 / 0.35	−10 dB midpoint 0.007 (clean) · centroid 0.026 (impaired) · 99 % midpoint 0.010 / 0.044 · peak 0.091 / 0.124
Bandwidth (relative error)	Noise-crossing ‡: 0.077 / 0.49	Noise-crossing ‡: 0.19 / 0.88	−26 dB 0.15 / 0.25 · 99 % occupied 0.18 / 0.30 · −10 dB 0.26 / 0.44 · −3 dB 0.52 / 0.76
SNR (error in dB)	Peak-PSD: 0.45 / 3.8 (bias −0.15)	Peak-PSD: 2.1 / 13.5 (bias −0.13)	In-band PSD 2.1–3.1 (bias −1.6 … −3.3) · M2M4 9.5 / 19.5

‡ Reproduces TorchSig's own labelling rule, so it partly measures the label itself. The best physical bandwidth measure is −26 dB (0.15 / 0.25).

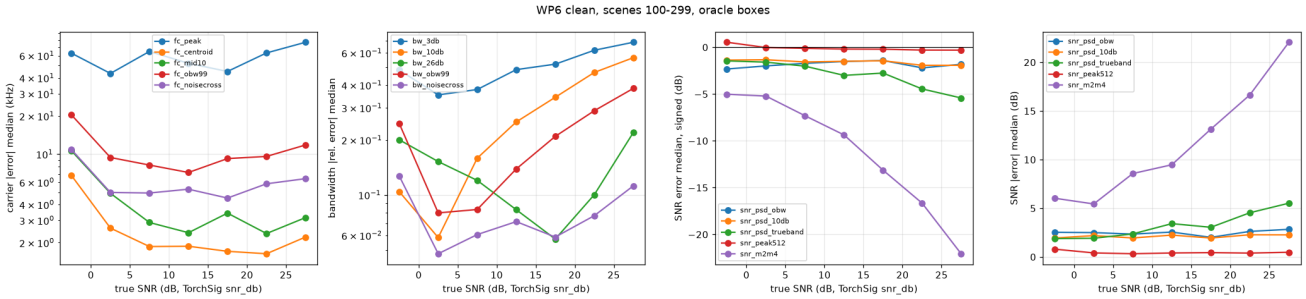


Figure 5. Clean scenes: carrier, bandwidth and SNR error vs SNR for every estimator.

Symbol rate

Synthetic RRC-shaped PSK/QAM (roll-off 0.2–0.5) and GMSK, 3–20 samples per symbol, 4,096 samples per frame. All estimators look for a cyclic spectral line at the symbol rate. Success = within 2 % of the true rate.

Estimator	All	BPSK	QPSK	8PSK	16QAM	GMSK	0 dB	20 dB
Instantaneous frequency, smoothed	79	92	94	95	85	30	62	95
Instantaneous frequency	76	99	95	96	84	6	59	83
Squared envelope	67	69	88	90	83	2	44	72
4th-power envelope	46	34	61	62	70	2	29	48
Haar wavelet	36	44	45	43	45	3	2	79
Delay-and-multiply	36	26	50	47	56	1	20	40

Success rate in %. With the modulation known (from 5.1) and the best estimator chosen per class: 81 %.

Second, independent generator: TorchSig's own modulators. PSK/QAM symbol rate = nominal bandwidth; FSK symbol rate captured during generation; labels checked on clean rectangular BPSK. 3,000 frames, SRRC or rectangular pulses, 0–20 dB. Success rate in %:

Estimator	All	BPSK	QPSK	8PSK	16 QAM	64 QAM	2FSK	4FSK	GFSK	MSK	GMSK
Inst. frequency, smoothed	73	100	99	98	92	92	66	43	32	66	40
Inst. frequency	49	88	89	90	82	83	1	1	20	6	27
Haar wavelet	43	75	78	76	72	76	2	0	17	6	23
Squared envelope	21	37	46	43	39	33	1	1	4	3	2
4th-power envelope	13	15	27	26	29	25	1	1	2	3	2
Delay-and-multiply	11	25	22	21	19	20	1	5	0	1	0

Envelope methods fail on rectangular pulses (squared envelope 70 % SRRC vs 9 % rectangular), as theory predicts. FSK-family symbol rate is the open gap. On RadioML the estimator places all 200 high-SNR BPSK/QPSK/8PSK frames at 7.5–8 samples/symbol: RadioML 2018.01A uses 8 samples/symbol.

Findings

- **TorchSig's *snr_db* is a peak-PSD SNR, not in-band signal-to-noise power** (verified in torchsig/utils/dsp.py, lines 1465–1469). In-band estimators read 2–3 dB below it, and chirp/LFM/LoRa labels are 25–48 dB below their physical SNR. **An SNR regressor (job 6.3) trained on TorchSig labels learns this definition.**
- TorchSig's bandwidth label grows with SNR (about 2× wider above 25 dB than below 5 dB), so fixed x-dB bandwidths are biased in an SNR-dependent way.

- M2M4 works only on truly constant-envelope signals (FSK/GMSK/FM); TorchSig pulse-shapes PSK, so even there it reads 7 dB low.
- GMSK symbol rate is the hard case: its frequency changes only at bit transitions, so it needs high SNR (95 % at 20 dB with smoothing).

5. Waveform-specific and auxiliary methods

These methods need a known waveform or work on top of a detector, so each is tested where its assumption holds, after first checking it against theory or a synthetic case with a known answer.

Method	Test	Result	Check
Matched filter (known 256-sample preamble) vs energy	Synthetic bursts, 4,096-sample frames, Pfa 10^{-3} per frame	Pd 90 % at **−10 dB** vs −4 dB for energy (**6 dB gain**); exact burst timing (energy ~260 samples off). With frequency offset ±0.002 cycles/sample: −8 dB	Theory −10.6 dB: passed
OFDM cyclic-prefix detector (Nfft 64, CP 16) vs energy	Synthetic OFDM, Pfa 10^{-2}, noise power known or ±1 dB uncertain	Known noise: energy −12 dB, CP −8 dB. **±1 dB noise uncertainty: energy false-alarm rate 0.43 (target 0.01); CP keeps 0.01** and still reaches 90 % at −8 dB (−6 dB blind over 4 FFT sizes)	Energy theory −12.4 dB: passed; SNR wall as in Tandra & Sahai: passed
MDL / AIC signal counting (single channel, 16-sample embedding)	0–4 tones in noise, 1,024 samples	**MDL: 90 % correct from −12 dB per tone, 100 % on noise only, no over-count at +10 dB.** AIC never reaches 90 %: 71 % on noise only, over-counts 32 % at +10 dB	MDL consistent, AIC over-estimates, as theory says: passed
CUSUM time-edge refinement of energy boxes	TorchSig scenes 100–299	Box F1 clean 67.4 → **68.1**, IoU 0.81 → 0.83; impaired 53.0 → **53.9**, IoU 0.79 → 0.82	Synthetic boxes: time-edge error 10.5 → 0 frames: passed
Wavelet band-edge refinement (Tian & Giannakis)	TorchSig scenes 100–299	Box F1 clean 67.4 → 53.1, impaired 53.0 → 44.7: **worse against TorchSig labels**	Synthetic: edge error 6 → 1 bin: passed. Refined boxes are 0.71× the truth width: the wavelet finds the physical edge, TorchSig labels reach to 0.5 dB above noise (definition mismatch, not a bug)
Hough transform for chirp / LFM / LoRa	TorchSig scenes 100–299, chirp-family emitters	Chirp recall at IoU ≥ 0.5: Hough alone 27 % (impaired) / 41 % (clean) vs energy segmentation 69 % / 78 %; **energy + Hough 77 % / 84 %**, with 1.4–1.6 extra false boxes per scene	Synthetic chirps: 18/20 correct, 0 false boxes on 20 non-chirp masks: passed. Tuned on dev scenes. With the line-thinness rule on (held-out): Hough alone 8 % / 6 %, energy + Hough 73 % / 81 %, only 0.10–0.17 false boxes per scene

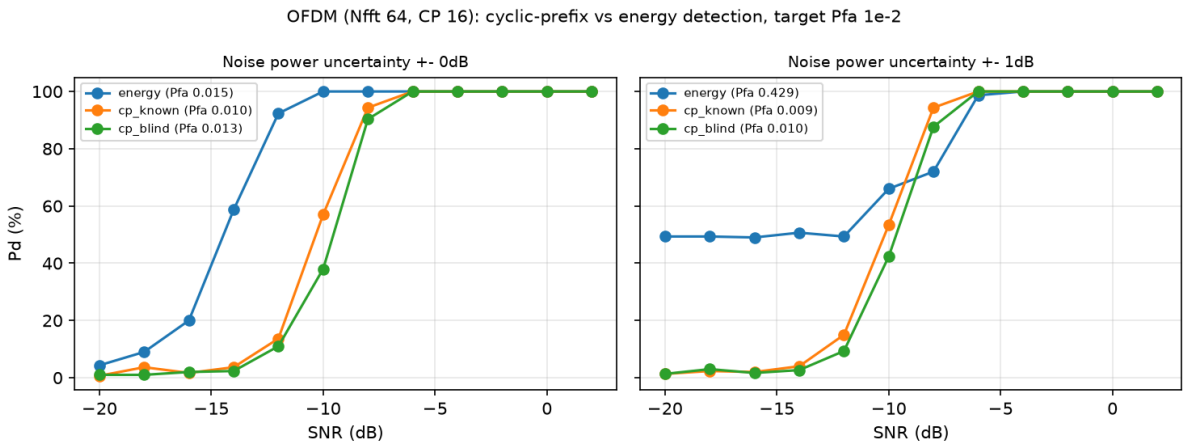


Figure 6. OFDM: cyclic-prefix vs energy detection with known (left) and ±1 dB uncertain (right) noise power. Measured false-alarm rates in the legend.

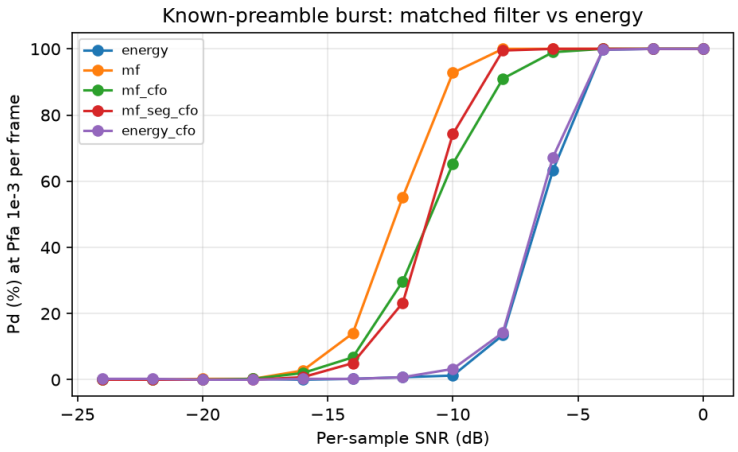


Figure 7. Known-preamble burst: matched filter (with and without frequency offset) vs energy detection.

6. Further method families (from the literature catalogue)

Added after enumerating the survey taxonomies (appendix). Each method passed a known-answer or theory check before benchmarking, and the key numbers were re-run independently (section 7).

Identification

Method (RadioML, boosted trees)	7 cls: all / SNR ≥ 10	24 cls: all / SNR ≥ 10	Note
19 features (reference)	71.4 / 99.95	39.2 / 62.8	
+ 17 extended and cyclic cumulants	71.4 / 99.97	39.7 / 63.5	no real gain
+ 24 Haar wavelet features	72.4 / 99.97	41.3 / 64.6	small real gain; reproduced exactly
+ 32 Hilbert–Huang (EMD) features	71.4 / 99.9	–	no gain; alone 48.9 / 67.8
Naive Bayes / QDA / per-class GMM (all SNRs)	64.0 / 70.1 / 70.8	28.4 / 35.5 / 37.5	below boosted trees

5 digital classes, symbol rate known: EM constellation classifier 95.2 % (RadioML) / 95.1 % (TorchSig) at SNR ≥ 10 dB, level with ALRT (93.3 / 95.6); Kolmogorov–Smirnov classifier 76.1 / 82.2 (weak with ~120 symbols).

Detection

Method	Result
Spectral correlation density (FAM)	Box F1 16.2 % impaired / 24.2 % clean: best block detector on impaired scenes, but 0–5 dB recall 1–5 %, ~4 s per scene
Covariance absolute value, autocorrelation, entropy	Box F1 6–8 % impaired (false-alarm rate ~0.3: the receiver filter colours the noise and breaks whiteness tests) / 37–39 % clean
Chirps (synthetic): spectrogram cell / line sum / PWVD / SPWVD / FrFT-dechirp	50 % detection at −13.4 / −20.7 / −21.7 / −21.4 / **−23.1 dB** (theory ≈ −23.7 dB)
S-transform vs STFT (64-sample bursts)	No gain: 90 % at 0 dB vs −4 dB for a burst-matched STFT (its window shrinks with frequency)
Burst onset: Shiryaev–Roberts / CUSUM / GLR / Bayesian online / Page–Hinkley	Mean delay 70.9 / 72.1 / 103.5 / 110.8 / 170.5 frames at equal false-alarm run length (−15 dB); CUSUM within 1–4 % of Siegmund's theory

Characterization

Method	Result
Carrier: periodogram ML, Quinn, Kay, data-aided ML	Reach the Cramér–Rao bound above threshold (MSE/CRB 0.9–1.5)
Carrier: Jacobsen; MUSIC / ESPRIT	≈2.2× CRB; 3–50× CRB (never reach it)
SNR: data-aided ML / split-symbol / M2M4 / eigenvalue	0.9–1.1× / 1–2.8× / ≈2× / 1.2–3.8× their bounds; M2M4 fails for 16QAM above 10 dB
Symbol rate: vote of 12 estimators (background-normalised lines)	**73** → **90 %** overall; FSK family **49** → **83 %** (GFSK/GMSK still weak below 5 dB)
Blind likelihood classifier with the new symbol-rate step	RadioML **80.4** → **91.8 %** at SNR ≥ 10 dB; 85 % now reached at 8 dB (bound with known rate: 93.3 %)

Final additions

Item	Result
Azzouz–Nandi rule tree (one feature per node, thresholds learned)	61.2 % / **98.6 %** at SNR ≥ 10 (7 classes), 85 % from 6 dB (learned tree 99.4 %); a first version with two badly chosen node features gave 66.1 % and was fixed
Swami–Sadler hierarchical cumulant classifier (symbols, training-free)	49.4 % / 64.4 % at SNR ≥ 10 (5 digital classes); limited by the blind M2M4 noise estimate (biased for QAM): 88 % with the true noise level on clean frames
MSK/GMSK symbol rate: squared-signal line pair	GMSK at 0 dB 93 % (vote 5 %), MSK at 0 dB 100 % (vote 45 %); not valid for GFSK with random index: GFSK below 5 dB remains open
Hough thinness trade-off (held-out)	Rule 0.5: chirp recall 73 % / 81 % at 0.10–0.17 false boxes per scene; rule ≥ 2 or off: 77 % / 84 % at 1.4–1.5. Recommended: 0.5
Chirp benchmark	All six 50 %-detection SNRs and the sanity values re-read from the result files: match
Newly found identification families	Histogram-entropy features: 24 classes 62.8 → **70.3 %** at SNR ≥ 10 dB (re-run independently: 69.7 %), the biggest classical gain on 24 classes; bispectrum and S-transform help only on 7 classes at low SNR; fractal small; CvM ≈ KS, AD worse
Newly found detection families	None beats energy detection. Teager–Kaiser ≈ energy (54.8 vs 53.0 impaired) with half the false alarms (re-run independently); eigenvalue family 6–9 % impaired (coloured noise); Tracy–Widom thresholds miscalibrated on smoothed covariances; stochastic-resonance gain not reproduced
Real over-the-air data	One recording tested: see the next section

First test on real over-the-air data

Allen Telescope Array RFI survey (Zenodo 8242048, CC-BY-4.0), recording 15-38-39: ADALM-Pluto, 61.44 Msps, centre 753.52 MHz, 2.28 s. Ground truth from the author's description: LTE carriers at 742.5 MHz (5 MHz), 751 MHz (10 MHz), 763 MHz (10 MHz), confirmed in the spectrum below.

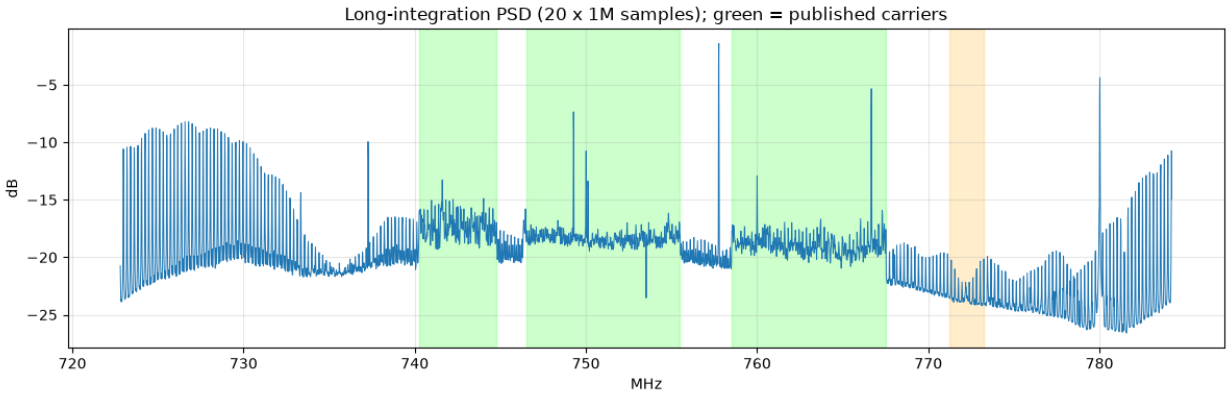


Figure 8. Real recording, long-integration PSD: the three LTE carriers (green) are only ~2 dB above the noise; a comb of receiver spurs covers both band edges.

Finding	Result
Time-frequency detectors, 4.3 ms segments	0 % of carriers detected by every detector: per-cell SNR below 0 dB
Frequency-only energy detection vs integration time	5 MHz carrier detected from 4.3 ms; 10 MHz carriers 21–24 % of bins at 4.3 ms → 47–56 % at 342 ms (the noise-floor estimate sits on wide carriers)
Noise model / false alarms	Quiet bins deviate from the Exp model (KS 0.021); flag rate 1–4 × 10^{-2} instead of 10^{-3}, mostly real receiver spurs
Carrier / bandwidth	742.5 MHz carrier: −97 kHz; 10 MHz carriers biased −1.2 / −1.8 MHz and 99 % OBW over-estimated by in-band spurs

On real data the bottleneck is weak signals and receiver spurs, not the detector choice: longer integration, spur notching and a wide-signal-safe noise floor come next.

7. Verification of delegated work

The WP5 classifier comparison and the WP6 estimators were written by helper agents. Their key results were re-derived independently:

- **WP5:** boosted trees reproduce the 5.1 run exactly (71.4 / 99.95 %); LDA (68.6 / 99.5 %) and k-NN (70.4 / 99.9 %) re-run from scratch on the same split give identical numbers.

- **WP6:** summary statistics recomputed from the raw per-emitter files match the agent's report except the impaired SNR 90th percentile (13.5 dB, not 10.5 dB; corrected in this report). A from-scratch re-implementation of the centroid carrier and peak-PSD SNR estimators gives 0.005 × bandwidth and 0.52 dB median error on clean scenes (agent: 0.004 and 0.45 dB). The TorchSig SNR definition was confirmed in its source code.
- **WP4:** every detector was checked to hit its target false-alarm rate on fresh white noise.
- **Section 6:** Haar-feature gain reproduced exactly; EM classifier 95.3 % on 300 random high-SNR RadioML frames (agent 95.2 %); symbol-rate vote 94.5 % on 200 random held-out frames with a fresh estimator computation (agent 89.9 % on all); blind ALRT 90.0 % on 150 random frames (agent 91.8 %); an independent ML tone-frequency estimator gives MSE/CRB 0.90–1.11; CAV and entropy detectors re-calibrated from scratch hit their target false-alarm rates, and a re-run on scenes 100–159 matches the agent table.

8. Not covered

Item	Reason
Real SDR recordings	Not recorded yet (jobs 2.1 / 2.4). Every number in this report is synthetic.
Matched filter / CP detector on the TorchSig scenes	TorchSig does not expose preambles or OFDM parameters per emitter, so these were tested on synthetic signals with known waveforms (section 5). Apply per standard once İLTAREN's signal list is fixed (M1).
Sub-Nyquist / compressive sensing, cooperative sensing, multi-antenna detection, MCMC classification	Not applicable: Nyquist-rate, single-receiver, single-antenna system; MCMC targets the same posterior as the tested EM classifier.
MDL / AIC for emitter counting in wideband scenes	With one channel, MDL/AIC count signal-subspace dimensions (tones), not emitters; a wideband emitter spans many dimensions. Tested on its valid use case (section 5).

9. Implications for the AI work

- **4.4 (YOLO detector)** must beat energy detection at 0–5 dB (40–71 % recall) and on impaired scenes (box F1 53 %).
- **5.2 / 5.3 (CNN, ResNet)** must beat 85 % at 2 dB on the 7-class set, or 63 % on all 24 classes; the likelihood bound suggests about 6 dB of headroom.
- **6.3 (ML regressor)** must beat 0.45 dB median SNR error and 0.4 % carrier error on clean data, and should be trained knowing the TorchSig SNR definition.
- Use the shared scoring (run_wp4.py, compare_classifiers.py, run_wp6.py) for the AI models so their numbers sit directly next to these.

Code and results: wp4_detection/, classical_amc/, wp6_characterization/, extra_methods/, classical_amc/extended/, detection_extended/, characterization_extended/, BASELINE.md and CATALOG.md in the project folder. Data: D:\data\radioml2018\ and D:\data\wideband\.

Appendix: catalogue of classical / statistical methods

Every method family named in the standard survey taxonomies (spectrum sensing: Axell, Leus & Larsson; Ahmad; modulation classification: Dobre et al.; Zhu & Nandi; plus the classical parameter-estimation, time-frequency and sequential-detection literature), with its status. T = tested (result given); N/A = not applicable, with the reason. Generated from CATALOG.md; section numbers refer to BASELINE.md.

A. Detection

Family	Method	Status	Where / result
Energy	Energy / radiometer (several noise-floor models)	T	BASELINE section 1: best detector, box F1 67 % clean / 53 % impaired
Energy	Multitaper (Thomson) energy	T	section 1
Energy	FCME, LAD (double threshold)	T	section 1
Energy	PSD occupancy (frequency only)	T	section 1
Energy	SNR wall under noise uncertainty	T	section 4: energy Pfa 0.43 instead of 0.01 at ±1 dB uncertainty
CFAR	CA, GO, SO, OS, 2-D CFAR	T	section 1, tuned guard/training sizes
Coherent	Matched filter / preamble correlator (plain, segmented for frequency offset)	T	section 4: 6 dB better than energy, matches theory
Feature: cyclostationary	2nd-order cyclic lines	T	section 1 (cyclo)
Feature: cyclostationary	Spectral correlation density (FAM/SSCA)	T	detection_extended/: FAM-based SCD per block; sanity: BPSK features at α = Rs and 2fc. Box F1 16.2 % impaired (best block detector, lowest Pfa 5.4e-2) / 24.2 % clean; recall 0–5 dB only 1–5 %; ~4 s per scene
Feature: autocorrelation	OFDM cyclic-prefix detector (known and blind FFT size)	T	section 4: robust to noise uncertainty
Feature: autocorrelation	Generic normalised-autocorrelation detector	T	detection_extended/: box F1 7.3 % impaired (Pfa 0.34) / 38.7 % clean. Impaired failure = receiver filter colours the noise (whiteness test fires); prewhitening helps impaired, hurts clean
Feature: covariance	Max/min eigenvalue (MME)	T	section 1
Feature: covariance	Covariance absolute value (CAV)	T	detection_extended/: box F1 6.6 % impaired (Pfa 0.32) / 36.8 % clean; same coloured-noise limitation. **Independent re-calibration on fresh noise hits target (1e-3 → 1.0e-3); subset re-run consistent**
Distribution	Anderson–Darling goodness of fit	T	section 1
Distribution	Spectral kurtosis	T	section 1
Information	Entropy-based detection	T	detection_extended/: spectral entropy F1 7.5 % / 39.1 %; histogram (Zhang) entropy 6.4 % / 12.5 %. **Independently re-run: consistent**
Multiband	Filter-bank / channelised detection	T	used inside the block detectors (16 sub-bands)
Multiband	Wavelet band-edge detection (Tian & Giannakis)	T	section 4: finds physical edges; worse vs TorchSig labels (definition mismatch)
Time–frequency	STFT spectrogram	T	basis of all WP4 detectors
Time–frequency	S-transform (Stockwell)	T	extra_methods/stransform_detect.py: no gain for 64-sample bursts (Pd 90 % at 0 dB vs −4 dB for a burst-matched STFT); its window shrinks with frequency
Time–frequency	Wigner–Ville / smoothed pseudo-WVD, fractional Fourier / Radon–Wigner (chirps)	T	detection_extended/ (synthetic chirps, 4,096 samples, Pfa 1e-2), SNR for Pd 50 %: radiometer −14.6, spectrogram cell −13.4, spectrogram line sum −20.7, PWVD −21.7, SPWVD −21.4, **FrFT/dechirp −23.1 dB** (theory ≈ −23.7 dB: checked); WVD energy 99 % on the IF line
Time–frequency	Hough transform on the spectrogram (chirps)	T	section 4: energy + Hough raises chirp recall to 77 % / 84 %
Sequential	CUSUM edge refinement	T	section 4
Sequential	CUSUM, Page–Hinkley, Shiryaev–Roberts, GLR, Bayesian online change-point (burst onset)	T	detection_extended/ (burst onset, ARL0 = 1000 frames, −15 dB): Shiryaev–Roberts 70.9 and CUSUM 72.1 frames mean delay (oracle-tuned), GLR 103.5, Bayesian online 110.8, mis-tuned CUSUM 168.8 (9.5 % miss), Page–Hinkley 170.5 (29.5 % miss). CUSUM ARL0 and delay within 3.7 % / 1.0 % of Siegmund's theory
Model order	MDL / AIC number of signals	T	section 4: MDL consistent, AIC over-counts
Sub-Nyquist	Compressive sensing (multicoset, MWC), alias-free sampling	N/A	Our SDRs sample the whole band at Nyquist; these methods matter only when the ADC cannot.
Cooperative	Soft / hard fusion across receivers	N/A	Single receiver in this project (possible future work with several SDRs).
Multi-antenna	Array GLRT / blind multi-antenna detectors, DOA	N/A	Single-channel receiver.

B. Identification (modulation classification)

Family	Method	Status	Where / result
Likelihood	ALRT, HLRT, ideal conditions	T	section 2: 85 % at 8 dB (upper bound)
Likelihood	ALRT, HLRT, practical (all parameters blind)	T	section 2: beats features when the symbol rate is right
Likelihood	Quasi-HLRT / GLRT variants	T	HLRT with noise level maximised (practical_alrt.py) is the GLRT over phase and noise
Distribution test	Kolmogorov–Smirnov classifier	T	classical_amc/extended/: 5 digital classes, symbol rate known: RadioML 60.2 % / 76.1 % (SNR ≥ 10), TorchSig 66.2 / 82.2, never 85 %; below features (87.1) and ALRT (93.3). Limitation, not bug: low test power with ~120 symbols, 64QAM mimics 16QAM
Feature: HOS	Cumulants C20…C80 (19-feature set)	T	section 2
Feature: HOS	Extended and cyclic cumulants	T	classical_amc/extended/: 17 extra (C6x, C8x, cyclic) cumulants, checked vs theory; +0.0 (7 cls) / +0.5 (24 cls) points: no real gain. Cyclic C40 survives frequency offset where plain C40 collapses
Feature: spectral / instantaneous	Azzouz–Nandi amplitude, phase, frequency features	T	inside the 19 features
Feature: power-law lines	x^2, x^4, x^8 spectral lines	T	inside the 19 features
Feature: wavelet	Haar wavelet features	T	classical_amc/extended/: 24 Haar features; boosted trees 71.4 → 72.4 % (7 cls), 39.2 → 41.3 % (24 cls; 62.8 → 64.6 % at SNR ≥ 10). **Re-run independently: identical**
Feature: Hilbert–Huang	EMD / IMF instantaneous-amplitude and frequency features	T	extra_methods/hht_features.py: EMD check passes (two sines separated, corr ≥ 0.999). Alone 48.9 % / 67.8 % (SNR ≥ 10), never 85 %; added to the 19 features 71.4 % vs 71.1 % (no real gain) at 18.7 ms/frame extra
Classifier	Decision tree (≈ Azzouz–Nandi thresholds), k-NN, SVM, LDA, random forest, boosted trees, nearest centroid	T	section 2
Classifier	Training-free cumulant matching (Swami–Sadler)	T	section 2
Classifier	Hierarchical cumulant classifier on recovered symbols (Swami–Sadler, training-free)	T	classical_amc/requirements_methods.py: noiseless check 100/100; RadioML 5 digital classes 49.4 % / 64.4 % (SNR ≥ 10); limited by the M2M4 noise estimate (88 % with true noise on clean frames)
Classifier	Rule tree, one feature per node, thresholds learned (Azzouz–Nandi)	T	classical_amc/requirements_methods.py: 61.2 % / 98.6 % (SNR ≥ 10), 85 % from 6 dB, vs 99.4 % learned tree (first version 66.1 % because of two badly chosen node features; fixed)
Classifier: Bayesian	Gaussian naive Bayes, QDA, per-class Gaussian mixtures	T	classical_amc/extended/: on the 19 features, naive Bayes 64.0 / QDA 70.1 / per-class GMM 70.8 % (7 cls); 28.4 / 35.5 / 37.5 % (24 cls): below boosted trees (71.4 / 39.2)
Blind	EM / Gaussian-mixture constellation classification	T	classical_amc/extended/: symbol rate known, RadioML 71.8 % / **95.2 %** (SNR ≥ 10), TorchSig 77.2 / 95.1, 85 % at 2 dB: level with ALRT, ~18× slower. **Spot-checked independently: 95.3 % on 300 frames**
Blind	MCMC (Gibbs) Bayesian classification	N/A	Targets the same posterior as EM at far higher cost; EM is the representative of this family.

C. Characterization

Quantity	Method	Status	Where / result
Carrier	PSD peak, centroid, x-dB midpoints, 99 % OBW midpoint	T	section 3
Carrier	Interpolated FFT (Jacobsen/Quinn), Kay, MUSIC, ESPRIT, M-th power, vs Cramér–Rao bound	T	characterization_extended/: periodogram ML, Quinn, Kay and data-aided ML reach the CRB above threshold (MSE/CRB 0.9–1.5); Jacobsen ≈2.2×; MUSIC/ESPRIT (M = 16/64) 3–50× (never reach it); QPSK 4th-power floors from self-noise. **Independent ML re-implementation: MSE/CRB 0.90–1.11**
Bandwidth	−3/−10/−26 dB, 99 % occupied (ITU-R SM.443), noise crossing	T	section 3
SNR	PSD-based (in-band, peak), M2M4	T	section 3
SNR	Split-symbol, data-aided / decision-directed ML, eigenvalue-based, vs CRB	T	characterization_extended/: CRB code reproduces the closed form to 0.1 %; data-aided ML 0.9–1.1× its bound; split-symbol 1–2.8×; M2M4 ≈2× (fails for 16QAM above 10 dB); eigenvalue 1.2–3.8× and biased low below 2 dB; kurtosis estimator shown identical to M2M4
Symbol rate	Envelope, 4th-power envelope, delay-multiply, Haar, instantaneous frequency (two test generators)	T	section 3
Symbol rate	FSK-specific estimators (closing the 32–66 % gap)	T	characterization_extended/: background-normalised line measurement + dev-tuned vote of 12 estimators: overall 72.7 → **89.9 %**, FSK family 49 → **83 %** (2FSK 95, 4FSK 97, MSK 88, GMSK 71, GFSK 64); GFSK/GMSK still weak below 5 dB. **Independent re-run on 200 random held-out frames: 94.5 %**
Symbol rate	MSK/GMSK squared-signal line pair	T	extra_methods/msk_line_pair.py: held-out GMSK at 0 dB 93 % (vote: 5 %), MSK at 0 dB 100 % (vote: 45 %); does not apply to GFSK with random index (25 %)
Symbol rate → classification	Blind likelihood classifier with improved symbol rate on RadioML	T	characterization_extended/: same 4,800 RadioML test frames, blind ALRT 80.4 → **91.8 %** at SNR ≥ 10 (sps-known bound 93.3), 85 % now at 8 dB; symbol rate within 2 % rises 70.6 → 92.7 %. **Independent re-run on 150 random frames: 90.0 %**
Number of signals	MDL / AIC	T	section 4

D. Discovery log (methods found after the first catalogue)

Found	Source	Status
Bispectrum features	arXiv 2604.10317 (review)	T: checks pass (Gaussian ≈ 0, coupled pair 0.997). 7 cls 71.4 → 73.1 %; 24 cls no gain (z −0.1)
Fractal-dimension features (Higuchi, Katz, box, multi-scale)	cumulants + fractal AMC paper; SPIE multi-scale fractal paper	T: Higuchi matches theory (line 1.000, white 2.000, Brownian 1.505). Small, not significant gains (z 1.6–1.7)
Information-theoretic features (Shannon / Rényi / permutation / sample entropy)	Gazi Univ. information-theoretic AMC paper	T: **24 classes 62.8 → 70.3 % at SNR ≥ 10** (8PSK +37, 16QAM +15 points), almost all from 8 cheap histogram entropies (0.4 ms/frame). **Independently re-run: 69.7 %**
S-transform features for classification	S-transform AMC paper	T: 7 cls 71.4 → 73.2 % (low-SNR gain); 24 cls small (z 1.9)
Cramér–von Mises and Anderson–Darling classifiers	EDF-test family (KS already tested)	T: match scipy to 1e-14, null calibrated. CvM ≈ KS (76.2 / 80.4 % at SNR ≥ 10); AD worse on RadioML (62.0 %: phase cut splits the BPSK cluster)
AGM, MED, John's test, eigenvalue moment ratio, condition number with Tracy–Widom thresholds	Huang & Fang 2015; Sedighi et al. 2017; small-sample sensing (arXiv 1404.5331)	T (detection_discovered/): box F1 6–9 % impaired / 35–37 % clean (coloured-noise limitation, like MME); Tracy–Widom thresholds miscalibrated on smoothed covariances (Pfa 5e-3–0.37 for 1e-3); EMR identical to John's test
Teager–Kaiser energy operator detector	MIT LL transient detection; arXiv 1504.08177	T: on the spectrogram ≈ energy detector (54.8 vs 53.0 impaired, 65.2 vs 67.4 clean) with half the false alarms on impaired scenes; weaker at 0–5 dB. **Independent re-run (scenes 100–159): 54.0 vs 53.7 / 68.8 vs 72.0**
Stochastic-resonance pre-treated energy detector	arXiv 0906.0739	T: paper's gain not reproduced; never better than plain energy detection (synthetic, 4,096 samples)
Hough trade-off	–	full held-out curve done: thinness rule 0.5 → 73 % / 81 % chirp recall at 0.10–0.17 false boxes/scene; ≥ 2 or off → 77 % / 84 % at 1.4–1.5